RESEARCHLY AI: AI-POWERED RESEARCH ASSISTANT AND COLLABORATION PLATFORM FOR UNIVERSITY STUDENTS

RESEARCH DATA COLLECTION AND MANAGEMENT MODULE: CORPUS, CATEGORISATION, SUMMARISATION AND OVERLAP ANALYSIS

Individual Thesis

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Institutional repositories store research output for deposit and display but not for analysis: metadata is incomplete and inconsistent, topics are author-assigned, summaries are absent and overlap is checked only per submission. This thesis presents the Research Data Collection and Management Module (Module 3) of Researchly AI, an integrated research platform built for SLIIT. The module works on a 4,219-record corpus from SLIIT's DSpace repository, pre-computes a 4,219 × 384 Sentence-BERT index shared with the other modules, audits corpus quality, categorises papers with a TF-IDF one-versus-rest classifier over 80 labels, produces faithful section-aware extractive summaries using centroid scoring and Maximal Marginal Relevance, and detects overlap through topic-level trend search and pairwise comparison combining semantic similarity (75%) with 4-gram Jaccard overlap (25%). The module follows Design Science Research and is deployed as a FastAPI micro-service behind an authenticated gateway with a Next.js interface.

The audit showed completeness 0.704 (no DOIs recorded), consistency 0.913 and a duplicate rate of 0.106. The summariser respected all length presets with about 89% correct section tags, and the comparator rated a paraphrased copy high risk (similarity 0.987, Jaccard 0.937) and unrelated text minimal. The classifier reached top-3 accuracy 0.692 but macro-F1 only 0.054, far below target, because author keywords form a sparse, CS-dominated label set. Of 14 validation cases, 11 passed, 2 were partial and 1 was blocked.

Keywords: research data management, data quality, multi-label classification, extractive summarisation, plagiarism analysis

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BART	Bidirectional and Auto-Regressive Transformer
BERT	Bidirectional Encoder Representations from Transformers
CPU	Central Processing Unit
DOI	Digital Object Identifier
DSR	Design Science Research
F1	Harmonic mean of precision and recall
FAIR	Findable, Accessible, Interoperable, Reusable
FR	Functional Requirement
GPU	Graphics Processing Unit
HNSW	Hierarchical Navigable Small World
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
LoRA	Low-Rank Adaptation
ML	Machine Learning
MMR	Maximal Marginal Relevance
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OvR	One-versus-Rest
PDF	Portable Document Format
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
RDA	Research and Digital Archive (SLIIT institutional repository)
REST	Representational State Transfer
RLS	Row-Level Security
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SBERT	Sentence-BERT
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SO	Specific Objective
SUS	System Usability Scale

Abbreviation	Description
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

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1 INTRODUCTION

1.1 Background and Overall Project Context

University students face a research workflow that has grown far faster than the tools supporting it. Global scholarly output rose from roughly 2.1 million articles in 2010 to more than 3.3 million in 2022 [1], and a final-year undergraduate must survey that literature, select a defensible topic, find a suitable supervisor, cite sources correctly, avoid overlap with earlier work and write a structured proposal, usually within a single academic year and with limited supervision time. Each of these tasks is served by a separate tool — a reference manager, a search engine, a similarity checker, a spreadsheet of supervisors — and none of these tools share data [2], [3].

The group project R26-IT-116, **Researchly AI — an AI-powered research assistant and collaboration platform for university students**, addresses this fragmentation by placing four intelligent modules behind one web application and one shared data substrate: Module 1 (Research Integrity and Compliance), Module 2 (Collaboration and Recommendation), Module 3 (Research Data Collection and Management) and Module 4 (Research Performance Analytics and Visualization). All four modules are grounded in a corpus of 4,219 records drawn from SLIIT's institutional research repository (RDA), of which 3,858 carry usable abstracts, so that every recommendation, gap, forecast or similarity score refers back to real prior SLIIT research rather than to generic web content. The platform is deployed as a Next.js front end on Vercel, an Express.js API gateway and four FastAPI micro-services, backed by Supabase PostgreSQL with the pgvector extension (Figure 1.1).

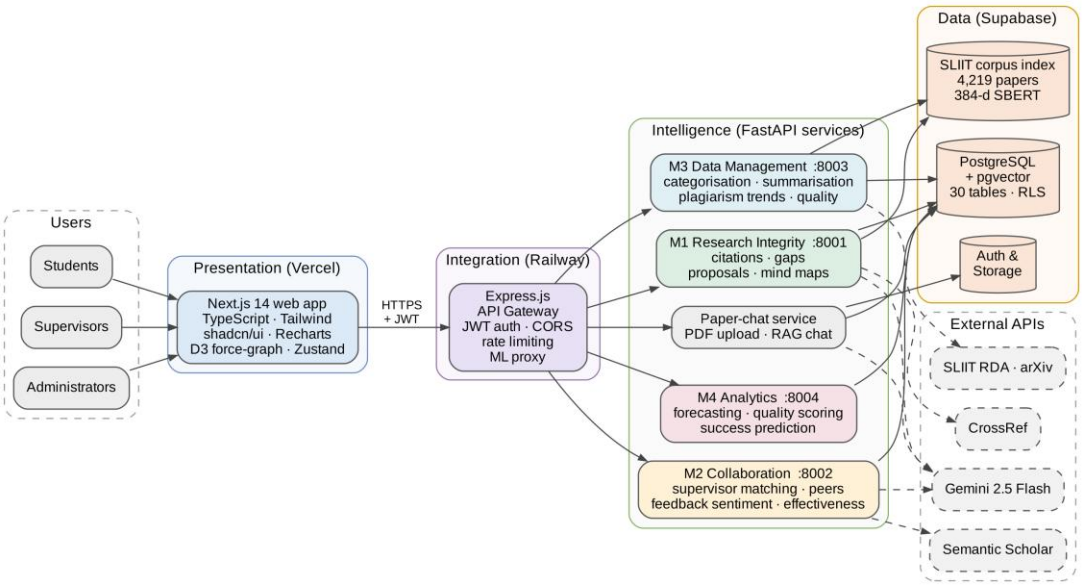


Figure 1.1: High-level architecture of the Researchly AI platform showing the presentation, integration, intelligence and data layers

This thesis reports the author's individual component, **Module 3: Research Data Collection and Management Module**. The common group report describes the integrated solution, the shared infrastructure and the end-to-end evaluation; this document concentrates on the research problem, design, implementation and validation of the individual component, and on how it exchanges data with the other three modules.

1.2 Individual Component Context

Module 3, the **Research Data Collection and Management Module**, is the data backbone of the platform. Every other module depends on a large, clean and consistently embedded corpus of research papers: Module 1 retrieves gaps and proposal exemplars from it, Module 2 shares its embedding model family, and Module 4 trains its forecasting and scoring models on it. Module 3 maintains that corpus and adds four user-facing capabilities, implemented as the FastAPI service `services/module3-data` (port 8003) and three pages under `/data-management`:

- **Topic categorisation** — multi-label classification of a paper into research categories with confidence scores and the most similar SLIIT papers.

- **Point-wise summarisation** — an extractive, section-aware summariser that turns a full paper (text or PDF) into a length-controlled bullet summary with a downloadable report.
- **Plagiarism-trend analysis** — topic-level trends of overlap over time, and pairwise comparison of two papers that combines semantic similarity with n-gram overlap and highlights the most similar sentence pairs.
- **Data-quality metrics** — completeness, consistency and duplicate-rate reporting over the corpus.

A shared in-memory **SLIIT paper index** — 4,219 records encoded once into a 4,219 × 384 embedding matrix — is reused by categorisation, plagiarism-trend search and, through the shared model files, Module 1.

1.3 Problem Statement

Universities generate large volumes of research output but keep it in repositories designed for deposit and display rather than analysis. Records carry inconsistent, author-chosen keywords, may lack identifiers, and are duplicated across collections; there is no systematic topic categorisation, no summary that helps a student decide whether to read a paper, and no institutional view of where similar work is being repeated [4], [5]. Students therefore spend hours skimming abstracts, administrators cannot see topic coverage, and similarity checking happens only per submission, never across the institution's history [2], [6].

The problem addressed by this component is: **the institution's research output is not available as a clean, categorised, summarised and similarity-aware corpus that students, staff and other system components can query.**

1.4 Research Gap

Table 1.1: Research gaps addressed by Module 3

ID	Gap	Evidence
G1	Institutional repositories store metadata but are not maintained as analysis-ready, quality-audited corpora.	RDM review [4]; FAIR [5]; data quality [7]
G2	Topic labels are author-assigned and inconsistent; automatic multi-label categorisation of institutional output is uncommon.	[8], [9]

ID	Gap	Evidence
G3	Abstractive summarisers risk hallucination and need GPU and domain adaptation; students need faithful summaries.	[10], [11], [12]
G4	Plagiarism tools compare one submission at a time; institutional overlap trends are not visible.	[2], [6], [13]

1.5 Aim and Objectives

Aim. To design, implement and evaluate a Research Data Collection and Management Module that collects, organises, categorises, summarises and quality-assures institutional research data and exposes it as a shared, similarity-aware corpus for the Researchly AI platform.

- **SO1** — Develop multi-label topic categorisation of research papers (proposal target: accuracy above 92%, macro-F1 at least 0.80).
- **SO2** — Build a plagiarism-trend analyser that detects overlap and topic repetition across institutional projects and compares two papers directly.
- **SO3** — Create a research summariser that produces faithful, structured summaries of papers.
- **SO4** — Implement data-quality assurance measuring completeness, consistency and duplication (target compliance 95%).
- **SO5** — Integrate the corpus and services with the other modules.

1.6 Research Questions

- **RQ1:** How complete, consistent and duplicate-free is the institutional corpus drawn from the SLIIT repository, and what does this imply for downstream models?
- **RQ2:** How well does a classical TF-IDF multi-label classifier categorise institutional papers across disciplines, and where does it fail?
- **RQ3:** Can an extractive, embedding-based summariser and a hybrid semantic–lexical comparator provide faithful summaries and reliable overlap detection on CPU-only infrastructure?

1.7 Scope, Limitations and Assumptions

In scope: metadata and abstracts in the SLIIT RDA corpus; English text; PDF and pasted-text summarisation and comparison; topic buckets for trend search; corpus-level quality metrics. **Out of scope:** full texts behind publisher paywalls and multilingual papers. **Assumptions:** repository metadata is broadly accurate; abstracts summarise their papers; corpus duplicates are rare enough not to dominate statistics.

1.8 Individual Contribution

The author designed and implemented the Module 3 service: the categorisation, summariser, plagiarism-trend and quality routers, the services `topic_classifier.py`, `extractive_summarizer.py` (302 lines), `plagiarism_analyzer.py` (436 lines) and `paper_index.py`, the training scripts for the topic classifier and plagiarism-trend index, the pre-computation of the 4,219-paper embedding matrix that removed out-of-memory failures at start-up, migration `004_module3_tables.sql`, and the three data-management pages of the web application.

1.9 Thesis Structure

Chapter 2 reviews related work; Chapter 3 describes the methodology; Chapter 4 the requirements and design; Chapter 5 implementation, integration and testing; Chapter 6 the results; Chapter 7 the discussion; Chapter 8 commercialisation; and Chapter 9 the conclusion, following the SLIIT guidelines [14].

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

Searches in IEEE Xplore, ACM Digital Library, ScienceDirect, PLoS and Google Scholar (2003–2026) used *research data management*, *institutional repository*, *data quality dimensions*, *duplicate record detection*, *multi-label text classification of scientific articles*, *extractive summarisation*, *maximal marginal relevance*, *plagiarism trends* and *document resemblance*. Works were retained if they defined a quality measure, a classification or summarisation technique, or a similarity method used or rejected in this module.

2.2 Domain and Theoretical Foundations

2.2.1 Research data management and repositories

Perrier et al. reviewed research data management in academic institutions and found infrastructure and practice fragmented, with repositories focused on preservation rather than reuse [4]. The FAIR principles require data to be findable, accessible, interoperable and reusable, including rich metadata and persistent identifiers [5]. DSpace, the platform behind SLIIT RDA, describes items with Dublin Core metadata [15].

2.2.2 Data quality

Wang and Strong's framework treats data quality as fitness for use across intrinsic, contextual, representational and accessibility dimensions, including completeness and consistency [7]. Duplicate record detection is a long-standing problem with exact, fuzzy and learned approaches [16]; even simple normalised-title matching provides a lower bound on duplication.

2.2.3 Multi-label text classification

In multi-label classification each document may belong to several categories; problem-transformation methods such as binary relevance (one-versus-rest) train one classifier per label [9]. TF-IDF weighting [17] with linear classifiers remains a strong baseline, and Mustafa et al. classified research articles with Word2Vec features and similarity

thresholds [8]. Domain-pretrained encoders such as SciBERT improve scientific-text classification [18] but require GPU training and inference for best results; topic models such as LDA [19] and BERTopic [20] discover rather than assign categories.

2.2.4 Summarisation

Extractive summarisation selects sentences from the source; TextRank ranks sentences by graph centrality [21], and Maximal Marginal Relevance (MMR) balances relevance against redundancy when selecting [22]. Abstractive models such as BART [10] and T5 [11] generate fluent summaries but can hallucinate unsupported statements [12]; long scientific documents require discourse-aware models [23]. ROUGE is the standard automatic measure [24].

2.2.5 Document similarity and plagiarism

Broder's resemblance measure — the Jaccard coefficient of shingle sets — underlies near-duplicate detection [25]. Alzahrani et al. classify plagiarism from verbatim copying to idea adoption and map detection methods to each [13]; Foltýnek et al. show that semantic methods are needed for paraphrase [2]. Curtis and Vardanega report changes in plagiarism behaviour over time [6], supporting the value of trend views. Sentence embeddings [26] provide the semantic signal.

2.3 Related Research

Table 2.1: Related research informing Module 3

Study	Approach	Relevance and limitation
Perrier et al. [4]	Scoping review of RDM	Repositories rarely analysis-ready; motivates a quality layer.
Wilkinson et al. [5]	FAIR principles	Metadata completeness and identifiers as quality criteria.
Wang and Strong [7]	Data-quality dimensions	Basis for completeness and consistency metrics.
Mustafa et al. [8]	Multi-label classification of research articles	Demonstrates feasibility; label imbalance a known challenge.
Beltagy et al. (SciBERT) [18]	Domain-pretrained encoder	Better accuracy, higher compute; planned but not deployed.
Carbonell and Goldstein (MMR) [22]	Diversity-aware selection	Core of the extractive summariser.

Study	Approach	Relevance and limitation
Lewis et al. (BART) [10]	Abstractive summarisation	Fluent but risk of hallucination; GPU cost.
Broder [25]	Shingle resemblance	Lexical tripwire for verbatim copying.
Foltýnek et al. [2]	Plagiarism detection review	Semantic signals required for paraphrase.

2.4 Existing Systems, Products and Techniques

Table 2.2: Comparison of existing products with Module 3

Product	Categorisation	Summarisation	Overlap analysis	Quality audit
DSpace repository (RDA)	Author keywords	No	No	No
Google Scholar / Semantic Scholar	Fields of study	TL;DR (some)	No	No
Turnitin	No	No	Per submission	No
Generic LLM chat	Ad hoc	Abstractive (unverified)	No	No
Module 3	Multi-label + related papers	Extractive, sectioned	Topic trends + pairwise	Completeness, consistency, duplicates

2.5 Gap Synthesis and Conceptual Framework

The review suggests that, for an institutional corpus served on CPU infrastructure, (i) quality must be measured before data is trusted, (ii) categorisation can start with a classical baseline but must be evaluated per discipline, (iii) summaries for academic use should be extractive so that every sentence is faithful, and (iv) overlap detection should combine semantic and lexical signals. Figure 2.1 shows where these services sit in the student journey.

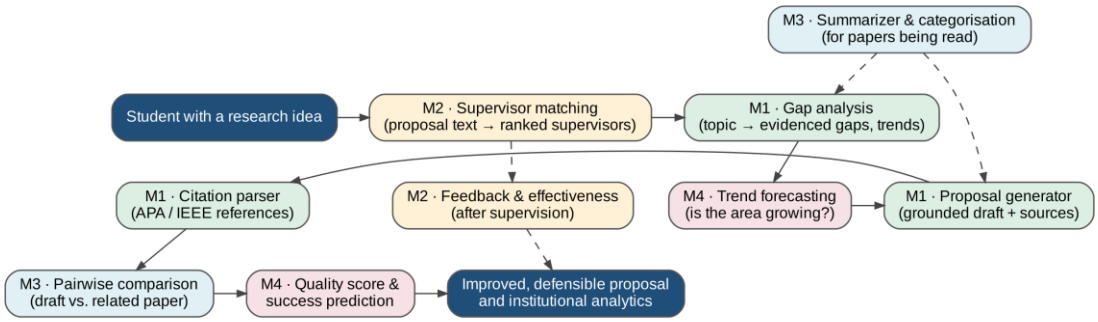


Figure 2.1: Student journey through the integrated platform; Module 3 supports reading (summaries, categories) and overlap checking

3 RESEARCH METHODOLOGY

3.1 Research Design

The component follows Design Science Research [27], [28]. Artefacts — the quality audit, classifier, summariser and comparator — were built iteratively and evaluated with corpus statistics, held-out classification metrics and functional validation through the running service.

3.2 Data, Participants and Study Materials

Table 3.1: Data and study materials used by Module 3

Material	Description	Use
SLIIT RDA corpus	4,219 records from the SLIIT RDA repository: id, title, handle, URL, authors, abstract, year, date, publication type, language, subjects	Shared corpus; quality audit
Embedding matrix	4,219 × 384 float32 SBERT embeddings of "title. abstract[:800]" (paper_embeddings.npy)	Related papers, trend search
Topic training data	1,382 papers with subject labels, 80 most frequent labels; 1,174 train / 208 validation	Topic classifier
Plagiarism-trend index	17 topic buckets, 27 topic-year records, 3,782 papers	Trend search
Test texts	Plant-disease abstract (10 sentences), lightly paraphrased copy, unrelated text, business and health abstracts	Functional validation

3.3 Experimental Procedure

1. Audit corpus quality with the completeness, consistency and duplicate-rate formulas of Section 4.4.4.
2. Build topic training data from subject keywords; train TF-IDF (30,000 features) with one-versus-rest logistic regression; evaluate on 208 held-out papers.
3. Pre-compute and persist the corpus embedding matrix; build topic buckets for trend search.
4. Implement the extractive summariser and the pairwise comparator; test them on controlled pairs.
5. Execute the 14-case functional validation on 28 August 2026.

3.4 Measures and Evaluation Metrics

Table 3.2: Evaluation measures for Module 3

Objective	Measure	Target
SO1	Micro- and macro-F1, top-1 and top-3 accuracy on held-out papers; behaviour per discipline	Macro-F1 ≥ 0.80
SO2	Document similarity, n-gram Jaccard and risk level for identical and unrelated pairs	Correct extremes
SO3	Compression ratio; section-tag correctness; faithfulness (sentences verbatim)	Length presets respected
SO4	Completeness, consistency, duplicate rate	Compliance ≥ 95%
SO5	Shared index used by other modules; routes reachable	Functional

3.5 Data Analysis

Quality metrics are computed over all records. Classification metrics follow standard multi-label definitions [9]; top-k accuracy counts a prediction correct if any gold label appears in the k highest-probability labels. Summaries were checked sentence by sentence against the source. Validation outcomes are classified PASS/PARTIAL/FAIL/BLOCKED.

3.6 Ethics, Privacy, Security and Responsible AI

- The corpus consists only of publicly available bibliographic records and abstracts from SLIIT's open research repository; full texts uploaded by users are stored in a private Supabase Storage bucket protected by Row-Level Security [29].
- Personal data (names, e-mail addresses, research interests, ratings) are processed in line with the principles of Sri Lanka's Personal Data Protection Act No. 9 of 2022 — purpose limitation, data minimisation and access control [30].
- Every request to an intelligence service passes through the gateway, which verifies a Supabase-issued JSON Web Token [31] and applies rate limiting (600 requests per minute globally and 200 per minute for ML routes).

- Generative components (Gemini) are used only where interpretation is genuinely required and every retrieval-based output is traceable to a cited SLIIT paper, which reduces the risk of hallucinated content [12].
- Summaries are extractive, so no statement is attributed to a paper that the paper does not contain.
- Overlap results are presented as similarity evidence with matched sentence pairs, not as findings of misconduct.

3.7 Reproducibility and Research Artefacts

The raw corpus (`data/papers_raw_sliit.json`), embedding matrix, training scripts (`training/prepare_topic_data.py`, `train_topic_classifier.py`, `prepare_plagiarism_corpus.py`, `train_plagiarism_analyzer.py`), model metadata and the validation harness (`docs/module-reports/_test_module3.py`) are versioned in the repository.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1: Functional requirements of Module 3

ID	Requirement
FR1	Categorise a paper into multiple categories with confidences and return related SLIIT papers; optionally write categories back to the paper.
FR2	Summarise pasted text or a PDF at four length presets, grouped by section, with an HTML report.
FR3	Search topic-level plagiarism trends and compare two texts or PDFs with a risk level and matched sentence pairs.
FR4	Report corpus completeness, consistency, duplicate rate and per-source counts.

Table 4.2: Non-functional requirements of Module 3

ID	Category	Requirement
NFR1	Performance	Summaries within seconds on CPU; related-paper search under 100 ms after warm-up.
NFR2	Reliability	No out-of-memory failure at start-up; graceful fallbacks for empty tables.
NFR3	Faithfulness	Summary sentences must be verbatim from the source.
NFR4	Security	JWT-authenticated routes; PDFs processed in memory.
NFR5	Maintainability	Feature logic isolated in service modules; models rebuilt by scripts.

4.2 Component Architecture within the Complete Solution

Figure 4.1 shows the component. The gateway route file `module3.routes.ts` (282 lines, the largest of the four) handles multipart PDF pass-through for the summariser and comparator and applies a 150-second timeout because the first request after a cold start may trigger a one-time corpus encode. Four routers delegate to the classifier, summariser, plagiarism analyser and quality aggregation; the shared paper index serves related-paper retrieval.

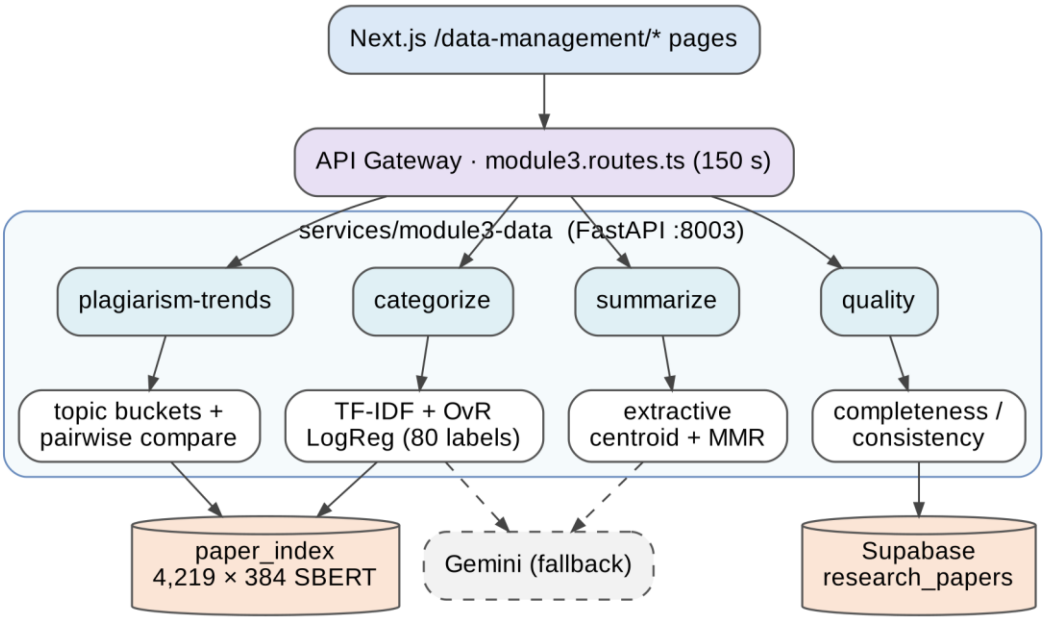


Figure 4.1: Component architecture of the Research Data Collection and Management Module

4.3 Design Alternatives and Decisions

Table 4.3: Design decisions and rationale for Module 3

Concern	Alternative considered	Decision	Rationale
Categorisation	Fine-tuned SciBERT, BCE loss [18]	TF-IDF + one-vs-rest logistic regression [17], [9]	Trains in seconds; calibrated probabilities; no GPU.
Summarisation	BART-large-CNN / Flan-T5 + LoRA [10], [11], [32]	Extractive SBERT centroid + lead bias + MMR [22]	Zero hallucination; ~80 ms vs ~5 s per paper; no fine-tuned weights needed.
Topic discovery	BERTopic (UMAP + HDBSCAN) [20]	Pre-computed topic buckets	Deterministic, fast trend search.
Pairwise overlap	Embeddings only	0.75 SBERT + 0.25 4-gram Jaccard	Paraphrase plus verbatim tripwire [25], [2].
Corpus index	Encode at start-up	Pre-computed .npy matrix, preloaded with a thread lock	Removed ~90 s cold encode and out-of-memory crashes.

4.4 Detailed Design

4.4.1 Topic categorisation

The classifier bundle (`classifier.pkl`, 19.5 MB) contains a fitted TF-IDF vectoriser (30,000 features), a one-versus-rest logistic-regression classifier and the

ordered list of 80 labels. Input text is vectorised and `predict_proba` gives one probability per label; if the estimator lacks probabilities, decision scores are passed through a sigmoid. Labels are sorted by probability: the top-k are returned for a "did you mean" view and those above the threshold (default 0.3; 0.2 in the UI) become the predicted categories. Related papers are retrieved from the shared index. If a paper ID is supplied, categories are written back to `research_papers.keywords` for future retrieval and filtering.

4.4.2 Extractive summariser

Figure 4.2 shows the summariser. Sentences of 20–600 characters are kept (at most 80). Each sentence is embedded and the normalised mean embedding forms a centroid representing the document's topical core. A base score combines three signals:

$$\text{base}(s) = 0.55 \cdot \cos(s, \text{centroid}) + 0.20 \cdot \text{position}(s) + 0.10 \cdot \text{length}(s)$$

where *position* is 1.0 in the first 20% of the document, 0.7 or 0.5 in the last 10% or 25% (papers restate key points at the end) and decays linearly otherwise, and *length* is 1.0 for 60–250 characters with a linear penalty outside. Sentences are then selected greedily by MMR [22]:

$$\text{next} = \operatorname{argmax}_{s \notin S} \left[\text{base}(s) - 0.15 \cdot \max_{t \in S} \cos(s, t) \right]$$

until the preset target (quick 5, standard 8–10, detailed 12–15, extensive 18–20 points) is reached. Each selected sentence is tagged by six ordered regex families (conclusion, limitations, results, methodology, objective, background) with a positional fallback, re-ordered into document order and grouped by section. A Gemini abstractive fallback exists only for pipeline failures.

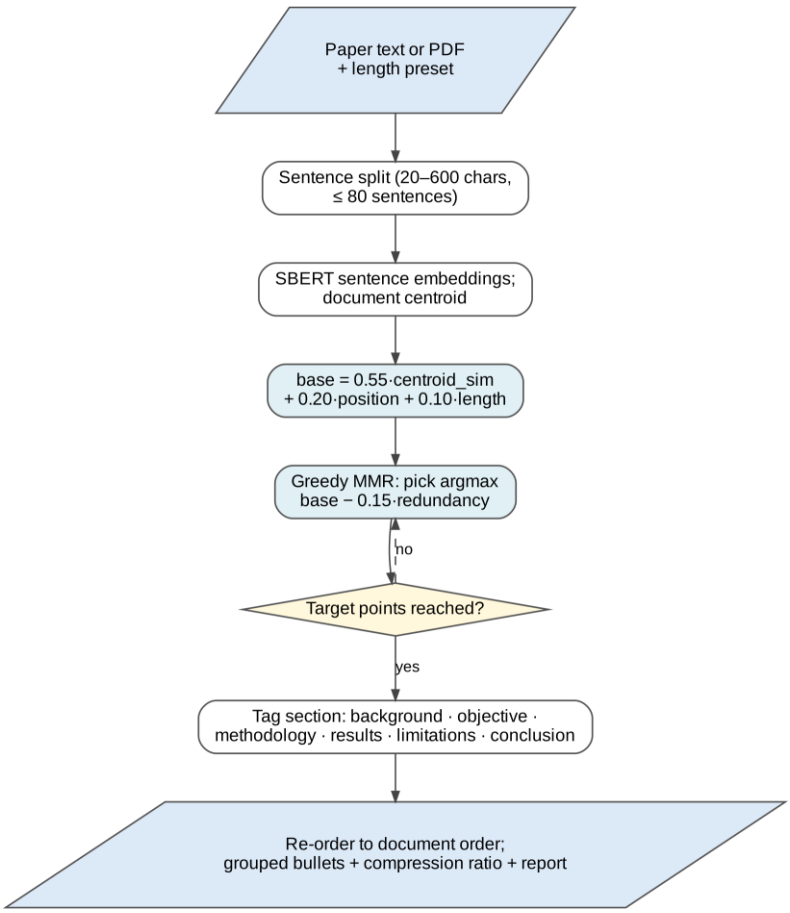


Figure 4.2: Extractive summarisation with centroid scoring and MMR selection

4.4.3 Plagiarism-trend search and pairwise comparison

Trend search embeds the query topic and scores it against pre-computed topic-bucket embeddings; buckets below a minimum similarity (default 0.25) are dropped, and each surviving bucket returns its per-year average, maximum and 95th-percentile similarity and high-similarity pairs, together with related SLIIT papers. Pairwise comparison (Figure 4.3) embeds both documents, computes 4-gram shingle sets [25] and the Jaccard coefficient plus one-sided overlap ratios, and computes a sentence similarity matrix (at most 60 sentences each) from which the top pairs above 0.5 are surfaced with `np.argpartition`:

$$\text{risk} = 0.75 \cdot \text{document_similarity} + 0.25 \cdot \text{jaccard_4gram}$$

Risk is *high* at 0.80 or above, *medium* at 0.60, *low* at 0.30 and *minimal* otherwise. Semantic similarity is weighted higher because paraphrase is the more common form of academic plagiarism [2]; n-grams remain a tripwire for verbatim copying.

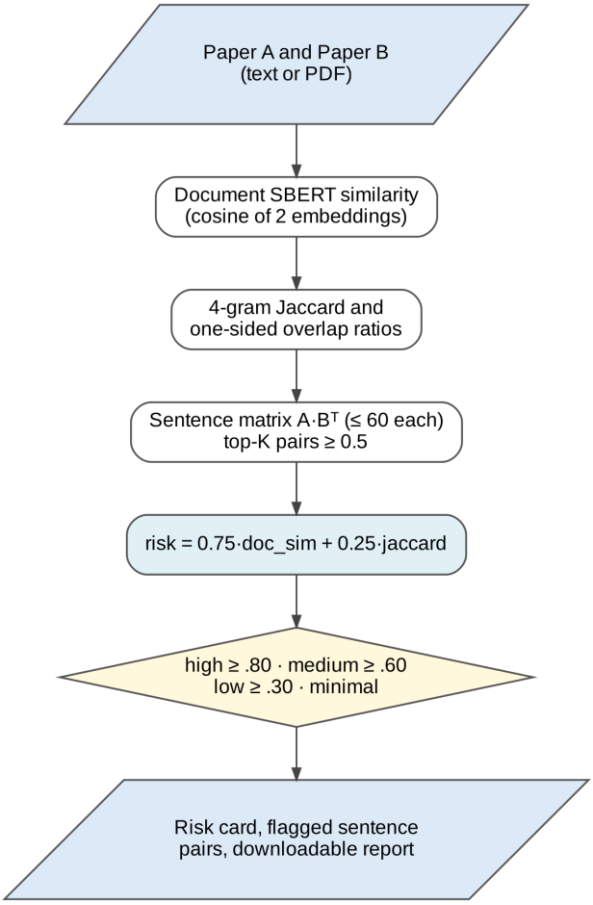


Figure 4.3: Pairwise comparison combining semantic and lexical overlap

4.4.4 Data-quality metrics

$$\text{completeness} = (n_abstract \geq 50 + n_doi + n_year + n_title \geq 5) / (4 \cdot N)$$

$$\text{consistency} = n(1950 \leq year \leq 2030 \wedge 50 \leq |abstract| \leq 10{,}000) / N; \quad duplicate_rate = 1 - n_unique_titles / N$$

These deliberately simple, transparent measures give the team visibility of corpus health before data is used for training or indexing [7], [16].

4.5 Standards and Professional Practice

- Metadata handling follows Dublin Core fields exposed by DSpace and the FAIR principles [5].

- Quality attributes follow ISO/IEC 25010 [33]; web security follows OWASP guidance [34].
- Attribution of every record to its repository handle.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technology stack

Layer	Technologies	Justification
Presentation	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, shadcn/ui, Recharts, D3 / react-force-graph-2d, Zustand, Axios	Server-side rendering, typed components, responsive layout and rich interactive charts in one framework [35]
Integration	Express.js API gateway (TypeScript), express-rate-limit, Supabase JWT verification, per-module proxy with 60 s / 150 s timeouts	Single entry point that authenticates once and isolates the front end from service locations [36]
Intelligence	Python 3.10+, FastAPI, sentence-transformers (all-MiniLM-L6-v2, fine-tuned), scikit-learn, XGBoost, statsmodels, PyMuPDF / pypdf / pdfplumber, Google Gemini 2.5 Flash client	Asynchronous typed micro-services; CPU-friendly models that fit the hosting budget [37], [38]
Data	Supabase PostgreSQL 15 + pgvector, Row-Level Security, Supabase Auth and Storage; pre-computed NumPy embedding matrices and pickled model bundles	Relational and vector search in one managed service, policy-based access control [39], [29]
DevOps	pnpm workspaces + Turborepo monorepo, Docker multi-stage images, GitHub Actions, Vercel, Railway, Azure Container Registry workflow, Playwright	Reproducible builds and automatic deployment on every push to main [40]

Module-specific dependencies include `scikit-learn` (TF-IDF, logistic regression), `sentence-transformers`, `numpy`, `pypdf` and `pdfplumber`.

5.2 Component Implementation

Table 5.2: Key Module 3 implementation files

File	Lines	Responsibility
app/services/plagiarism_analyzer.py	436	Trend-bucket search, pairwise comparison, HTML reports
app/routers/plagiarism_trends.py	387	Trend search, legacy trends, compare (text/PDF), reports
app/routers/summarizer.py	313	Text/PDF summarisation, presets, report, Gemini fallback
app/services/extractive_summarizer.py	302	Centroid scoring, MMR selection, section tagging
training/prepare_plagiarism_corpus.py	221	Topic buckets and per-year similarity statistics
app/services/paper_index.py	203	4,219 × 384 index, filters, related papers
web: data-management/plagiarism-trends/page.tsx	860	Trend search, text and PDF comparison UI

File	Lines	Responsibility
web: data-management/summarizer/page.tsx	393	Summariser UI with presets and report download
web: data-management/categorization/page.tsx	240	Categorisation UI with confidence bars and related papers

The paper index preloads the persisted matrix at start-up and scores queries with one matrix product, applying optional year and subject filters by masking rather than slicing:

```
sims = _EMBEDDINGS @ qvec  # cosine because both are normalised
mask = np.ones(sims.shape, dtype=bool)
...                         # year-range / subject filters set mask[i] =
False
sims_filtered = np.where(mask, sims, -1.0)
order = np.argsort(-sims_filtered)[: max(top_k, 1)]
```

Figures 5.1 to 5.6 show the implemented interface.

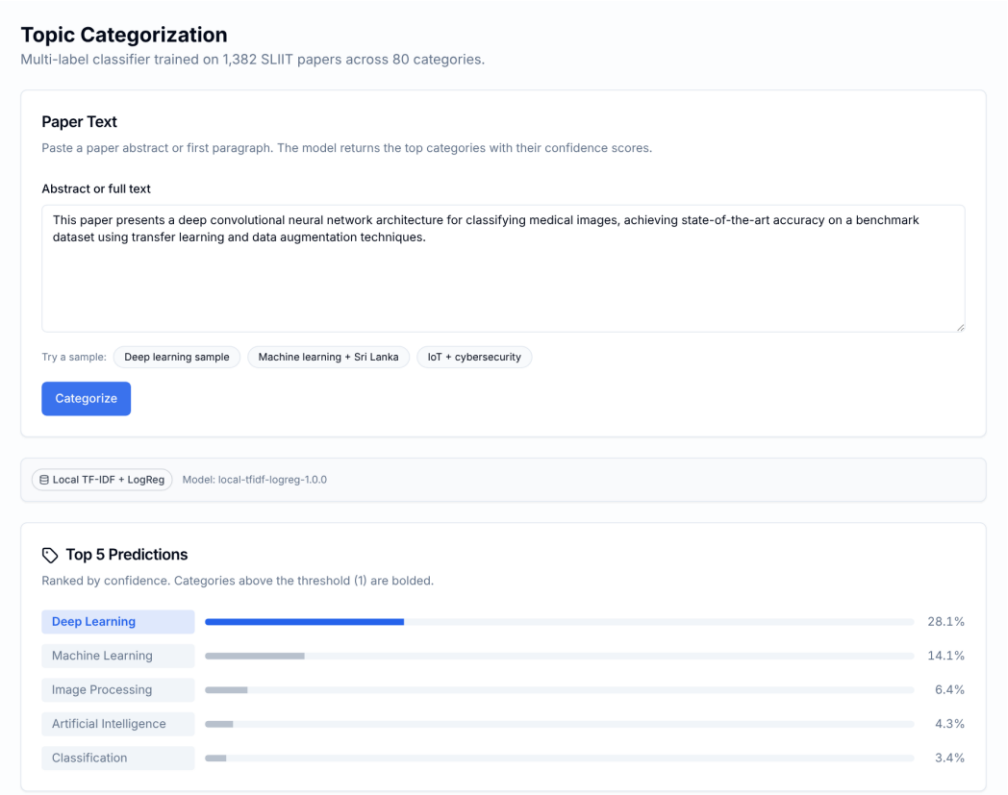


Figure 5.1: Topic Categorisation page: abstract input and top-5 predicted categories with confidence bars

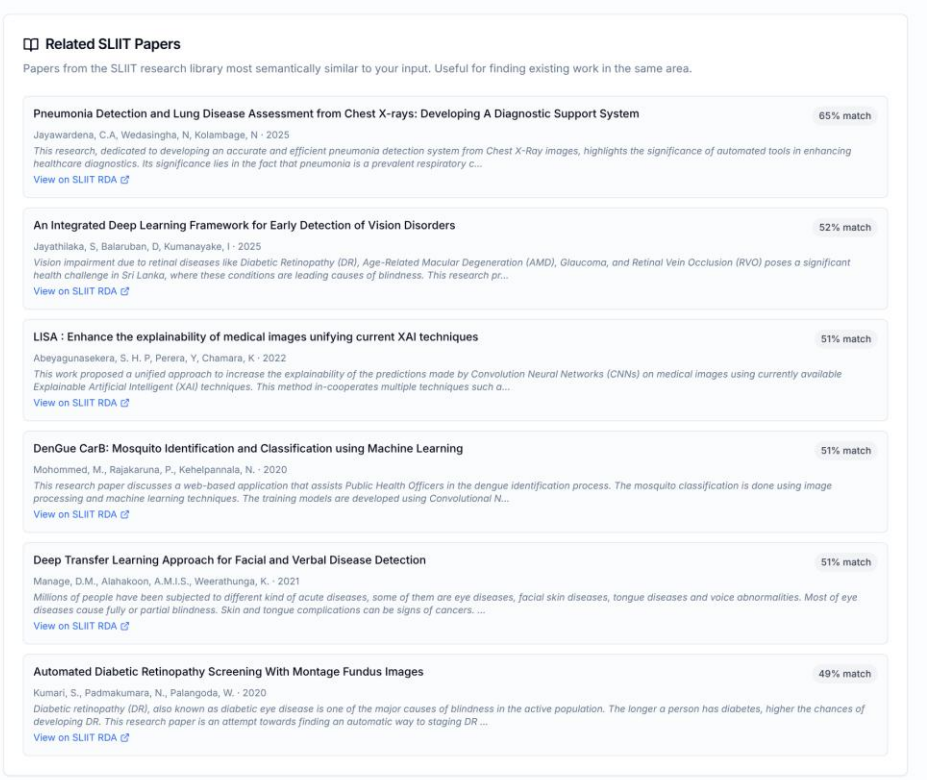


Figure 5.2: Related SLIIT papers retrieved for the categorised abstract, with match scores

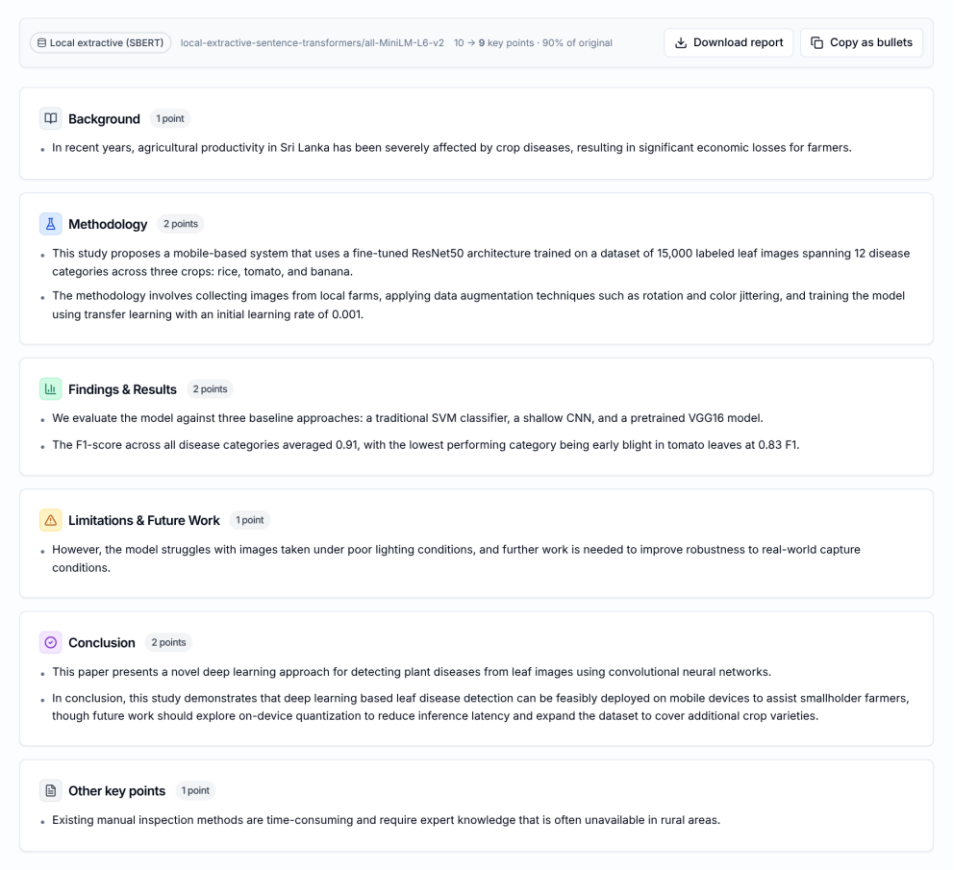


Figure 5.3: Section-grouped extractive summary (10 → 9 key points) with model badge and report download

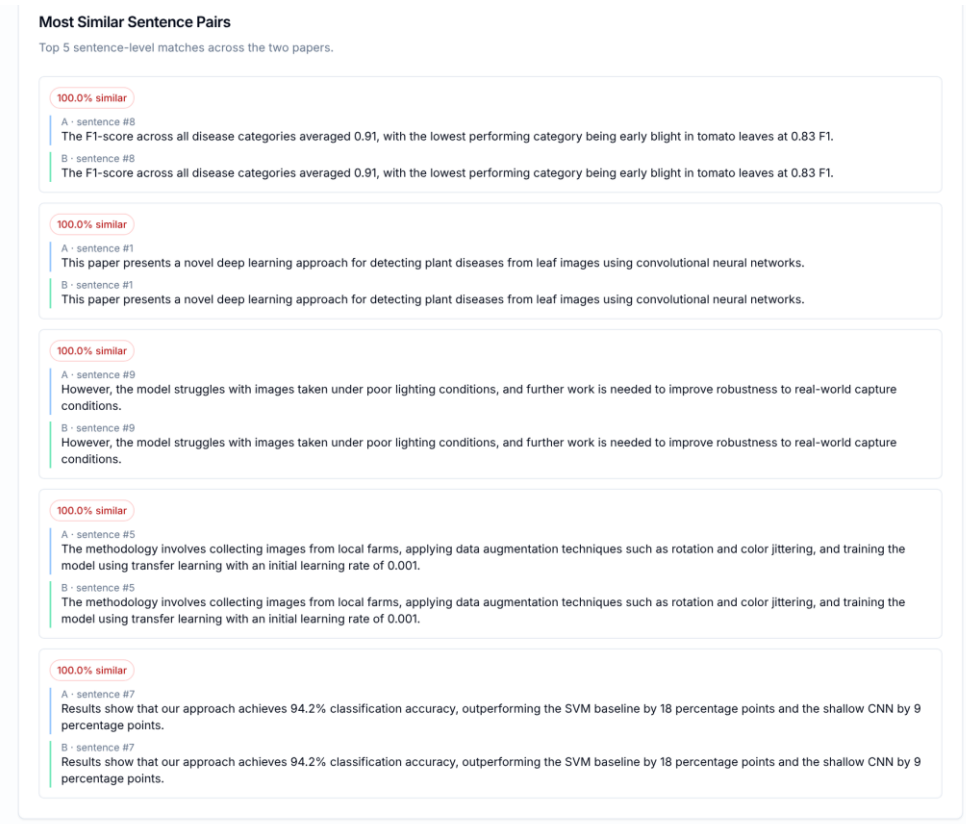


Figure 5.4: Plagiarism comparison result: HIGH risk with SBERT document similarity 98.7%, 4-gram Jaccard 93.7% and matched sentence pairs

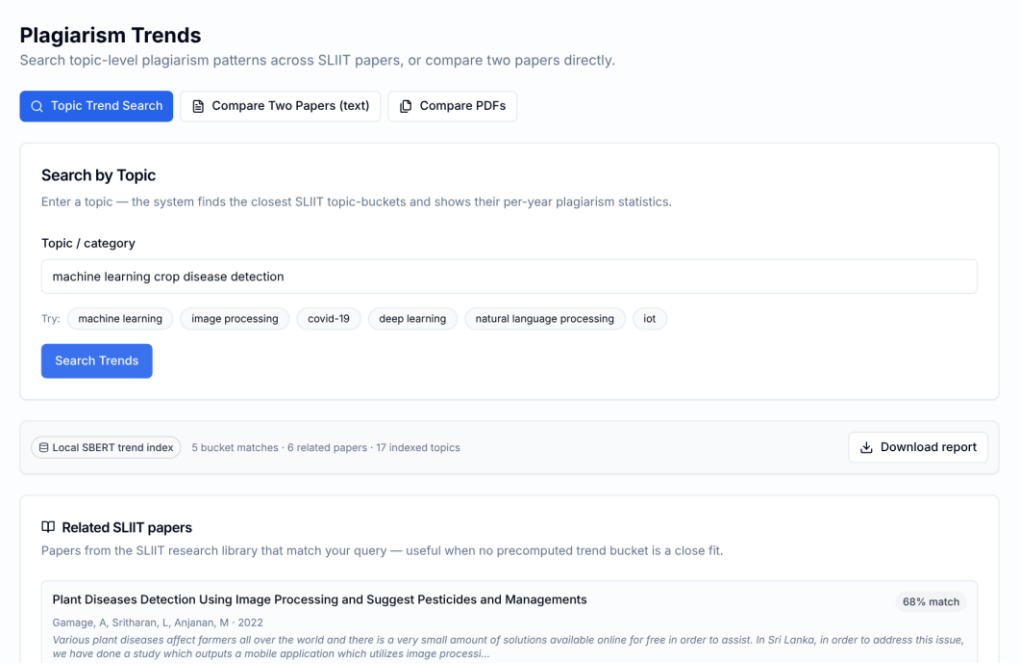


Figure 5.5: Topic-level plagiarism-trend search with related SLIIT papers

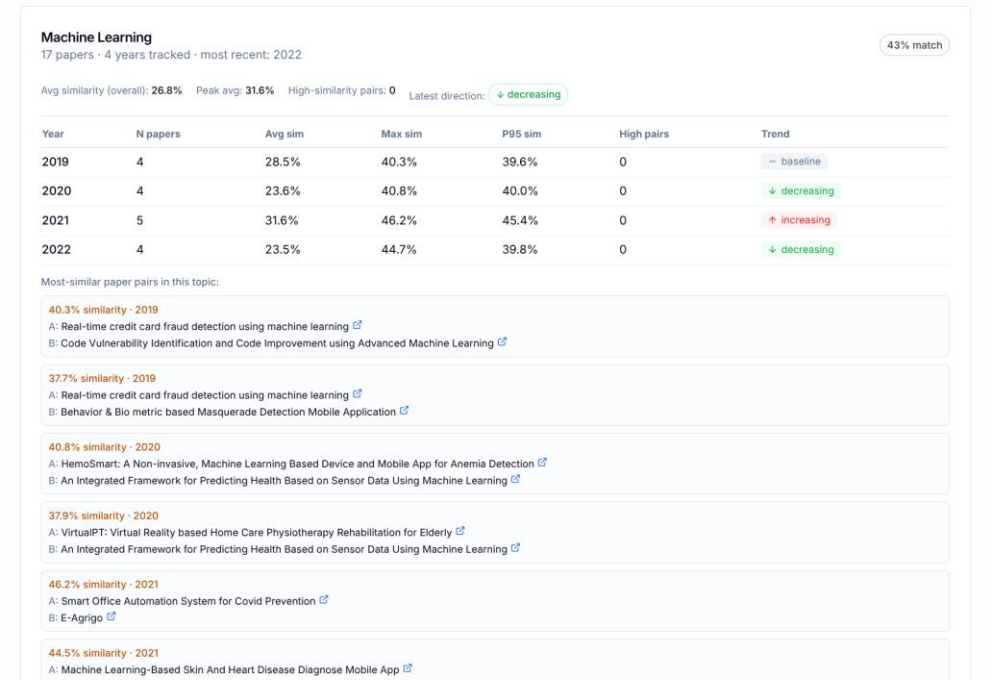


Figure 5.6: Per-year similarity statistics and most-similar paper pairs for the "Machine Learning" bucket

5.3 Integration with the Other Components

- **Module 1:** consumes the corpus, the derived citation/proposal datasets and the shared SBERT model.
- **Module 2:** shares the MiniLM embedding family; duplicate records observed in Semantic Scholar profile data corroborated Module 3's duplicate findings.
- **Module 4:** trains trend, quality and success models on the corpus.
- **Platform:** shared gateway, authentication and dashboard shell (Figure 5.7).

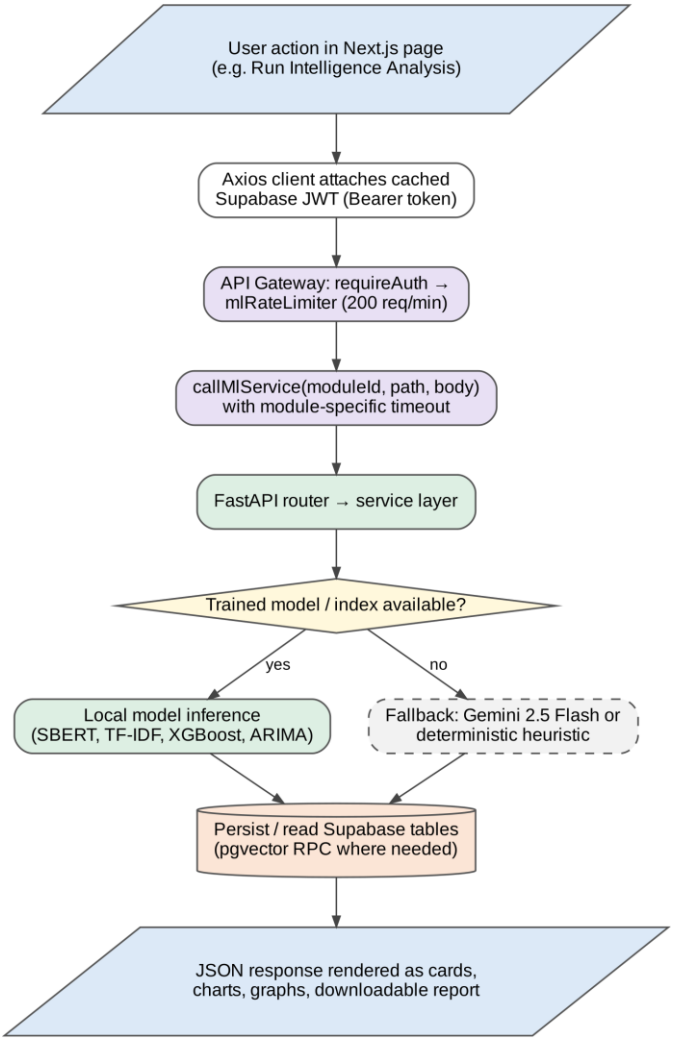


Figure 5.7: Request flow from a Next.js page through the gateway to a FastAPI service

5.4 Verification and Software Testing

Corpus-level checks are reported in Chapter 6. Functional validation executed 14 cases against the running service on 28 August 2026 (Table 5.3). The Playwright test `e2e/module3-summarize.spec.ts` covers the summariser page.

Table 5.3: Functional validation of Module 3 (28 August 2026)

ID	Scenario	Result	Evidence or root cause
M3-01	Categorise — CNN medical-imaging abstract	PASS	deep learning 0.28 (top); related papers on medical imaging
M3-02	Categorise — business abstract	PARTIAL	No category above 0.2; related papers correct
M3-03	Categorise — health abstract	PARTIAL	Top confidence 0.10; CS-dominated label set
M3-04	Classifier status	PASS	80 labels loaded

ID	Scenario	Result	Evidence or root cause
M3-05	Summarise — standard	PASS	9 of 10 sentences, ratio 0.904; ~89% correct section tags
M3-06	Summarise — quick	PASS	5 of 10 sentences, ratio 0.585
M3-07	Summarise — extensive	PASS	All 10 sentences (target exceeds input)
M3-08	Summariser status	PASS	Algorithm description accurate
M3-09	Trend search — ML crop disease	PASS	"machine learning" bucket (0.43) + related papers
M3-10	Legacy trends — empty table	PASS	Fell back to local corpus; exact-duplicate pairs noticed
M3-11	Compare — lightly paraphrased copy	PASS	doc 0.987, Jaccard 0.937 → high
M3-12	Compare — unrelated pair	PASS	doc −0.009, Jaccard 0 → minimal
M3-13	Plagiarism analyser status	PASS	17 topics loaded
M3-14	Data-quality endpoint	BLOCKED	Supabase table empty in test environment

5.5 Deployment and Operation

The service image bakes in the SBERT model and the pre-computed embedding matrix; an earlier deployment that encoded the corpus at start-up exceeded memory on Railway, which was resolved on 13 May 2026 by pre-computing `paper_embeddings.npy`, preloading it at start-up behind a thread lock and raising the gateway timeout for Module 3 to 150 seconds. Summariser and PDF-comparison latency was also reduced on the same day (Figure 5.8).

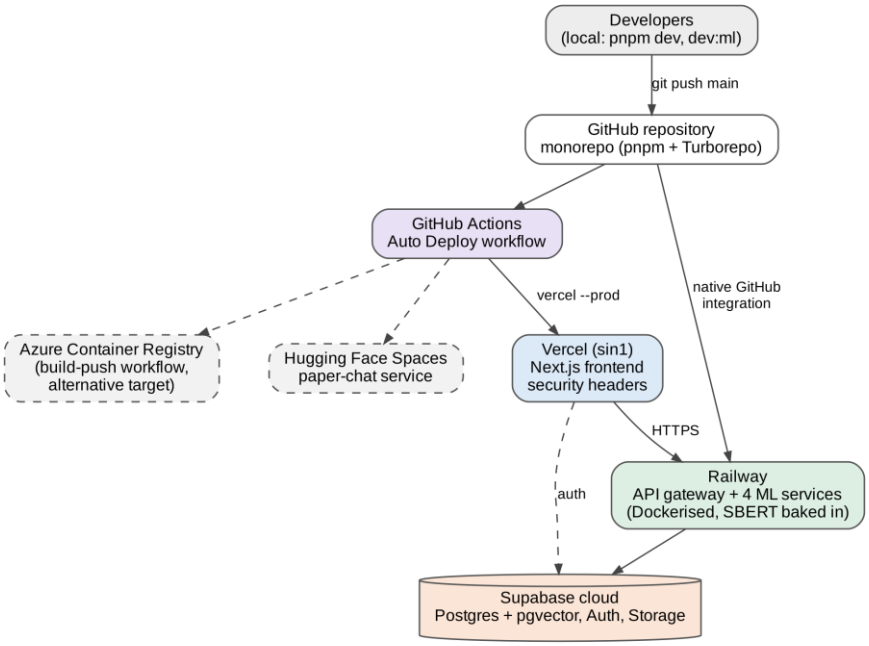


Figure 5.8: Continuous deployment pipeline of the platform

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 RQ1: Corpus quality

The corpus contains 4,219 records spanning 1986–2026, with strong growth after 2017 and peaks in 2021 (633 papers) and 2022 (575) (Figure 6.1). The 2026 count (34) reflects a partial year at collection time, an important caveat for any time-series model built on the corpus.

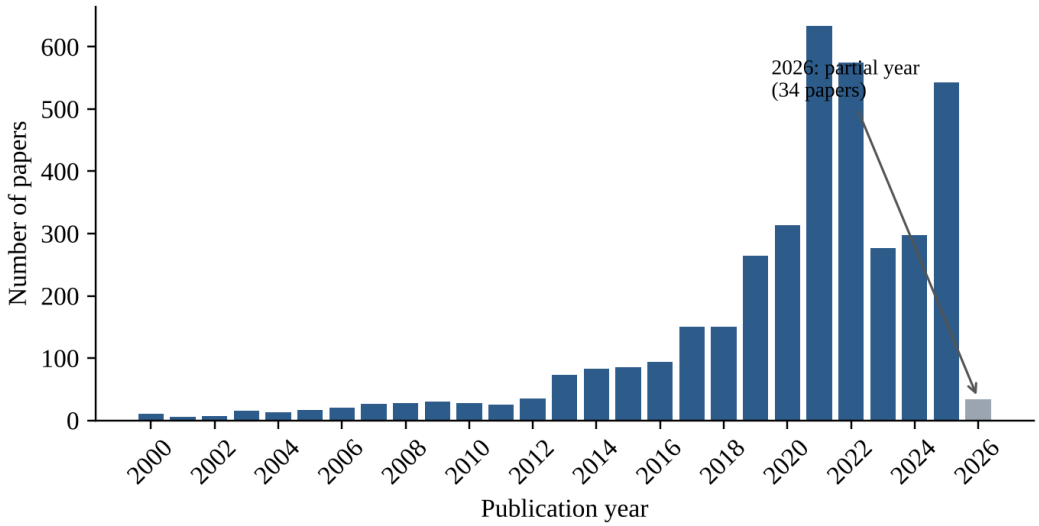


Figure 6.1: SLIIT corpus records by publication year (2000–2026)

Applying the quality formulas offline to all 4,219 records (Figure 6.2, Table 6.1) gives completeness 0.704, consistency 0.913 and a duplicate rate of 0.106. Completeness is dragged down entirely by the absence of DOIs (0%), which SLIIT RDA does not record; title (98.4%), year (91.7%) and abstract (91.4%) coverage are high. Ten per cent of records share a normalised title with another record, corroborating validation finding M3-10.

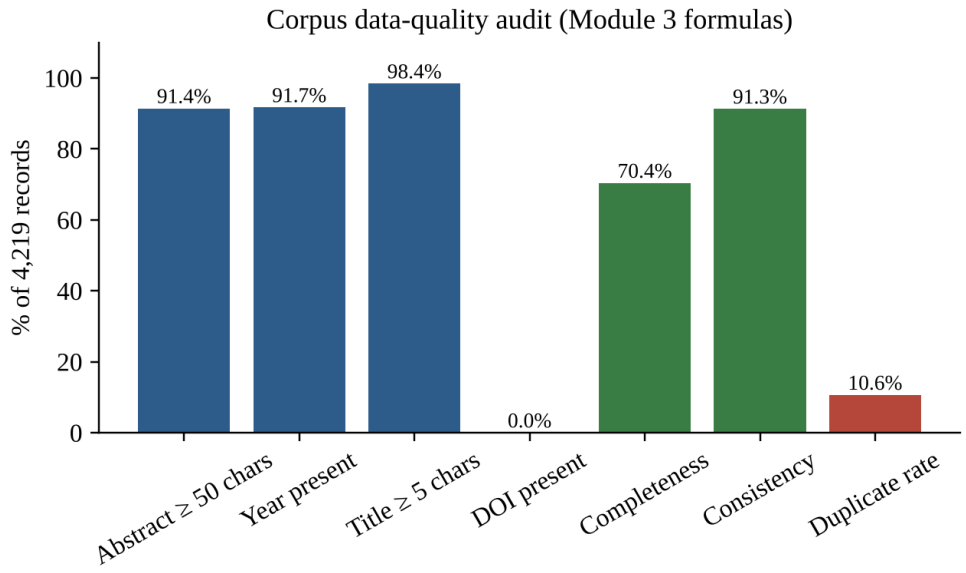


Figure 6.2: Corpus data-quality audit computed with the Module 3 formulas

Table 6.1: Corpus quality audit (N = 4,219)

Measure	Value	Target (SO4)
Records with abstract ≥ 50 characters	3,858 (91.4%)	—
Records with valid year	3,870 (91.7%)	—
Records with title ≥ 5 characters	4,153 (98.4%)	—
Records with DOI	0 (0.0%)	—
Completeness	0.704	≥ 0.95
Consistency	0.913	≥ 0.95
Duplicate rate (normalised titles)	0.106	Minimise
Median abstract length	1,326 characters	—
Mean authors per paper	3.39	—

6.1.2 RQ2: Topic categorisation

On 208 held-out papers the 80-label classifier achieved micro-F1 0.154, macro-F1 0.054, top-1 accuracy 0.481 and top-3 accuracy 0.692 (Figure 6.3), far below the proposal target of macro-F1 0.80. The label set is the 80 most frequent author keywords, dominated by computing terms (machine learning, deep learning, image processing) and including near-duplicates (*neural network* and *neural networks*, *iot* and *internet of things iot*), with many labels having very few examples; macro-F1 is therefore driven down by rare labels. Validation confirmed the pattern: a CNN medical-imaging abstract was categorised correctly, but business and health abstracts

produced no category above the threshold, even though the SBERT related-paper retrieval for the same inputs was highly relevant.

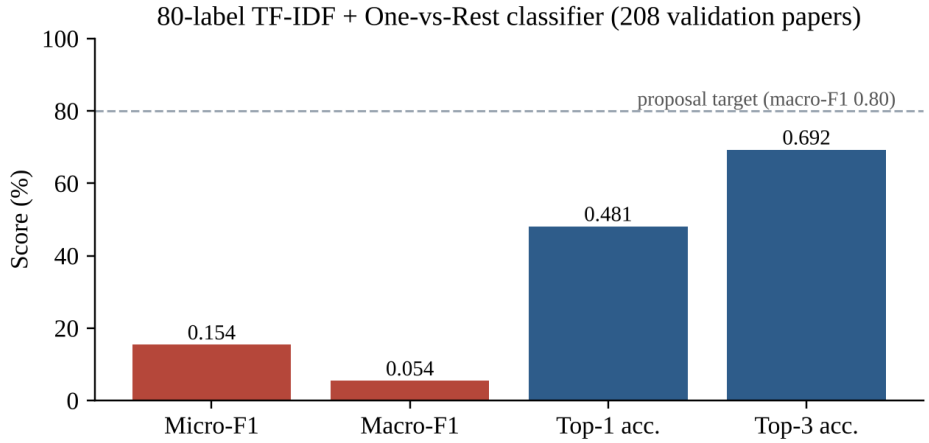


Figure 6.3: Held-out metrics of the 80-label TF-IDF topic classifier

6.1.3 RQ3: Summarisation and pairwise comparison

For a 10-sentence structured abstract, the standard preset selected 9 sentences (compression ratio 0.904), the quick preset 5 (0.585) and the extensive preset all 10, so length control behaved as specified; about 89% of section tags were correct, the main error being an opening "this paper presents…" sentence tagged as conclusion (Figure 5.3). Because every bullet is a source sentence, the summary is faithful by construction. Pairwise comparison was accurate at both extremes: a lightly paraphrased copy scored document similarity 0.987 and 4-gram Jaccard 0.937 (risk *high*), while unrelated text scored −0.009 and 0 (risk *minimal*).

6.2 Product and Technical Results

Table 6.2: Feature completion status of Module 3

Feature	Status	Notes
Topic categorisation + related papers	Partial	Classifier weak outside computing; retrieval strong
Extractive summariser + report	Complete	Four presets; PDF upload
Plagiarism-trend search + pairwise comparison	Complete	Text and PDF; HTML reports
Data-quality endpoint	Implemented	Validated offline; live table empty
SciBERT, BERTopic, BART models	Not deployed	Wrapper classes present

Non-functional evidence: the pre-computed index eliminated out-of-memory start-up failures and made related-paper search a single matrix product (about 30 ms); summarisation is reported at about 80 ms per paper on CPU versus about 5 s for a BART forward pass; and empty-table cases degrade gracefully (M3-10).

6.3 User and Stakeholder Validation

The supervisor reviewed and signed off the module validation report, identifying the summariser and comparator as strongly validated and the classifier's discipline bias and corpus duplicates as the priorities. A task-based student acceptance protocol is provided in Appendix D.

6.4 Integration Results

The corpus and embedding matrix are consumed by Modules 1 and 4, and all 14 Module 3 routes are reachable through the gateway with authentication and the extended timeout.

6.5 Summary of Findings

The module delivers an institutional corpus with high title, year and abstract coverage but no DOIs and about 10% duplication; faithful extractive summaries; and accurate overlap detection. Topic categorisation with a classical model over noisy author keywords did not meet its target. In validation, 11 of 14 cases passed, 2 were partial and 1 was blocked (Figure 6.4).

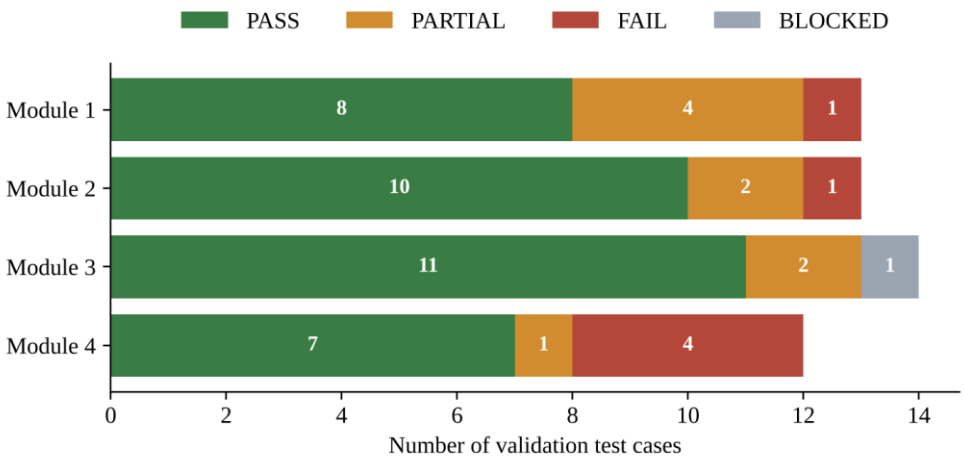


Figure 6.4: Validation outcomes per module (Module 3 in the third row)

7 DISCUSSION

7.1 Interpretation of Findings

The quality audit explains several behaviours observed elsewhere in the platform. Missing DOIs mean that identity across sources must rely on titles, and the 10.6% duplicate rate means that exact-duplicate pairs inflate plagiarism-trend maxima (M3-10). The partial 2026 year explains why Module 4's growth percentages were distorted. The classifier result is primarily a *label* problem: author keywords are sparse, overlapping and CS-dominated, so even a well-tuned model would struggle; the strong performance of SBERT related-paper retrieval on the same inputs shows that the representation, not the input text, is the bottleneck. The extractive summariser trades fluency for faithfulness, which is the right trade for academic use.

7.2 Comparison with Literature Baselines and Existing Solutions

Published multi-label classifiers of research articles achieve far higher scores on curated taxonomies [8], [18]; the present result is a reminder that label design dominates. Extractive MMR summarisation [22] is a classical approach that remains competitive in faithfulness with abstractive models [10], [11]. The hybrid comparator combines Broder's resemblance [25] with semantic similarity as recommended by plagiarism reviews [2], [13]. Unlike repository software [15], the module adds analysis on top of storage.

7.3 Research Contribution

- A reproducible quality audit of an institutional research corpus, quantifying completeness, consistency and duplication and their downstream effects.
- Evidence that author-keyword label sets are unsuitable targets for multi-label classification without curation.
- A faithful, CPU-efficient extractive summariser with section tagging and a hybrid semantic–lexical comparator validated at both similarity extremes.

7.4 Product Contribution

The institution gains a single, embedded and audited corpus that powers every module; students gain fast, faithful summaries and categories with related papers; and supervisors gain a way to compare two documents with explainable evidence.

7.5 Limitations and Threats to Validity

- **Classifier:** macro-F1 0.054 against a 0.80 target; business and health under-represented.
- **Duplicates:** about 10% of titles duplicated; not yet removed from the index.
- **Quality endpoint:** validated offline only, because the live table was empty.
- **Summariser evaluation:** one structured abstract and no ROUGE evaluation against reference summaries [24].
- **Coverage:** the corpus is a snapshot of the repository.
- **Planned models:** SciBERT, BERTopic and BART were not deployed, so proposal-level targets for those models are not tested.

7.6 Implications

For institutions, investment in metadata quality (DOIs, controlled vocabularies, de-duplication) yields more value for AI services than model complexity. For research, categorisation studies should report label-set construction alongside metrics. For students, faithful extractive summaries are a safer default than abstractive ones.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

Primary users are research administrators, library staff and department heads who need an analysable view of institutional output; **secondary users** are students and supervisors who summarise, categorise and compare papers. The module turns a deposit-oriented repository into a queryable research data service.

8.2 Market and Competitive Position

Repository platforms store records; global indexes cover external literature; similarity checkers compare single submissions (Table 2.2). The module is differentiated by combining quality audit, categorisation, faithful summarisation and institutional overlap analysis over the institution's own corpus, and by serving that corpus to the other platform modules.

8.3 Business and Sustainability Model

Table 8.1: Indicative revenue model (from the approved proposal)

Phase	Strategy	Indicative pricing
Year 1	Free for SLIIT departments; corpus onboarding and refinement	—
Year 2	Freemium: summaries free; analytics and comparison reports premium	LKR 600 per department per month
Year 3	Institutional licence with repository integration	LKR 200,000 per institution per year

Operating costs are low because indexing and summarisation run on CPU; the proposal budgeted LKR 131,000 for development resources.

8.4 Scalability, Intellectual Property and Compliance

- Exact search over 4,219 vectors is trivial; for $10^{5–10}6$ records, pgvector HNSW indexes would be used [41], [39].
- Audit scripts, summariser scoring and comparator weighting are the team's work; corpus metadata remains attributed to the source repository.

- Only public metadata is used; uploaded PDFs are private; compliance with repository terms and the PDPA [30] is required for external deployment.

8.5 Contribution to the Sustainable Development Goals

The platform contributes primarily to **SDG 4 (Quality Education)**, in particular target 4.3 (equal access to quality tertiary education) and target 4.4 (relevant skills), by giving every student — not only those with well-connected supervisors or paid tools — structured support for research. It also contributes to **SDG 9 (Industry, Innovation and Infrastructure)**, target 9.5 (enhance scientific research and upgrade technological capabilities), by turning an institutional repository into queryable research infrastructure [42]. Module 3 contributes most directly to SDG 9 by building research data infrastructure and to SDG 4 by making institutional knowledge discoverable and digestible for students.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Research Data Collection and Management Module of Researchly AI. It audited the institutional corpus that underpins the platform, provided faithful extractive summarisation and accurate hybrid overlap detection on CPU infrastructure, and exposed the limits of classifying papers by noisy author keywords. The quality audit made the corpus's strengths and weaknesses explicit, which in turn explained behaviours in the other modules.

9.2 Achievement of Objectives

Table 9.1: Achievement of objectives

Objective	Status	Evidence
SO1 Multi-label categorisation (macro-F1 ≥ 0.80)	Not achieved (0.054)	Section 6.1.2
SO2 Plagiarism-trend analyser and comparison	Achieved	Section 6.1.3; M3-09 to M3-12
SO3 Research summariser	Achieved (extractive)	Section 6.1.3; M3-05 to M3-07
SO4 Data-quality assurance (95% compliance)	Partly achieved (metrics implemented; corpus below target)	Section 6.1.1
SO5 Integration with other modules	Achieved	Section 6.4

9.3 Individual Research and Product Contributions

- Designed the corpus quality audit and quantified completeness, consistency and duplication.
- Implemented the extractive summariser, hybrid comparator, trend search and topic classifier with their interfaces.
- Resolved start-up memory failures by pre-computing and preloading the embedding matrix.

9.4 Future Directions

- Curate a controlled vocabulary of 15–25 discipline-balanced categories and retrain (or fine-tune SciBERT [18]); use SBERT embeddings as classifier features.
- De-duplicate the corpus (normalised titles plus embedding similarity) and enrich DOIs through CrossRef.
- Evaluate summaries with ROUGE [24] against human-written summaries and conduct the user study in Appendix D.
- Populate the live Supabase tables and re-run the data-quality validation.

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GLOSSARY

Term	Definition
Completeness	Share of required fields (abstract, DOI, year, title) present across records.
Consistency	Share of records whose year and abstract length fall within plausible ranges.
Centroid	The normalised mean of all sentence embeddings of a document.
MMR	A greedy selection rule that trades relevance against redundancy with already selected items.
Shingle	A contiguous sequence of n words used for lexical overlap.
Topic bucket	A pre-computed group of corpus papers with per-year similarity statistics.
Extractive summary	A summary built only from sentences copied verbatim from the source.

APPENDICES

Appendix A: Extended Results and Test Evidence

Table A.1: Fifteen most frequent subject keywords in the corpus

Keyword	Papers	Keyword	Papers
sri lanka	235	economic growth	30
machine learning	189	development	29
deep learning	74	iot	28
image processing	69	optimization	27
artificial intelligence	58	computer vision	26
covid-19	55	social media	24
natural language processing	46	e-learning	23
mobile application	22		

Recorded comparison output for validation case M3-11:

```
{ "document_similarity": 0.987, "ngram_jaccard": 0.937,
  "risk_level": "high", "flagged_pairs": [ { "similarity": 1.0, ... } ]
}
```

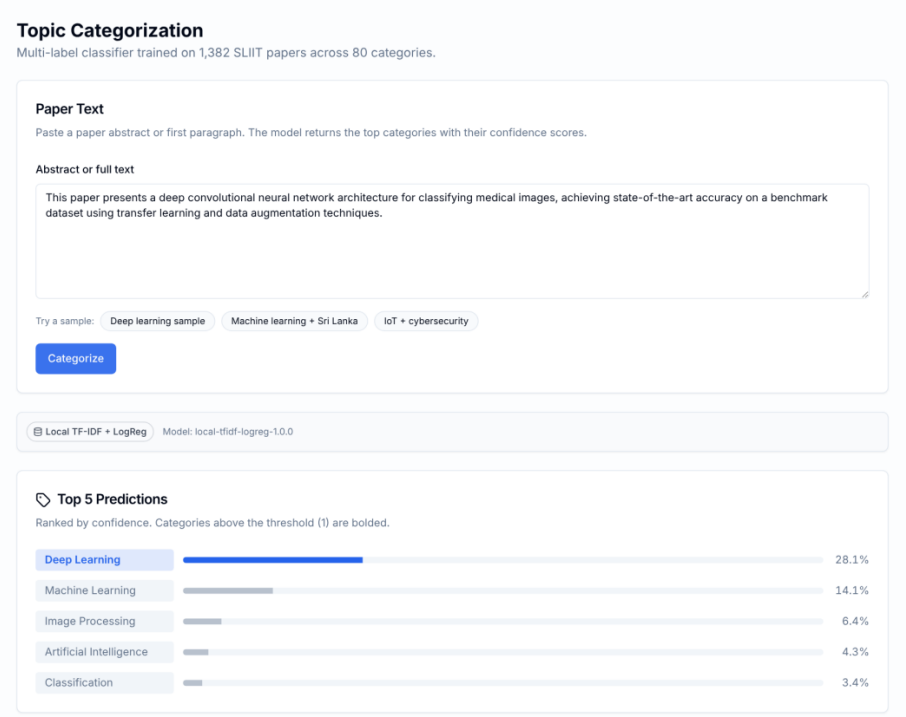


Figure A.1: Categorisation input form with sample abstracts (full view)

Appendix B: Academic Integrity and AI-Use Disclosure

Table B.1: AI-use disclosure

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic) — AI coding assistant	Debugging suggestions; execution of the module validation harness and drafting of the module report (28 Aug 2026)	Evidence inspected; supervisor sign-off obtained	Chapter 5.4, Chapter 6
Claude (Anthropic)	Drafting and formatting this thesis from repository evidence; offline corpus audit, diagrams and charts	All statistics recomputed from repository files and cross-checked	All chapters
Google Gemini 2.5 Flash	Runtime fallback for categorisation and summarisation	Outputs treated as untrusted	Chapter 4

Appendix C: Additional Interface Screenshots

Screenshots were captured from the production build of the Researchly AI front end rendering responses recorded from the live Module 3 service during the 28 August 2026 validation run.

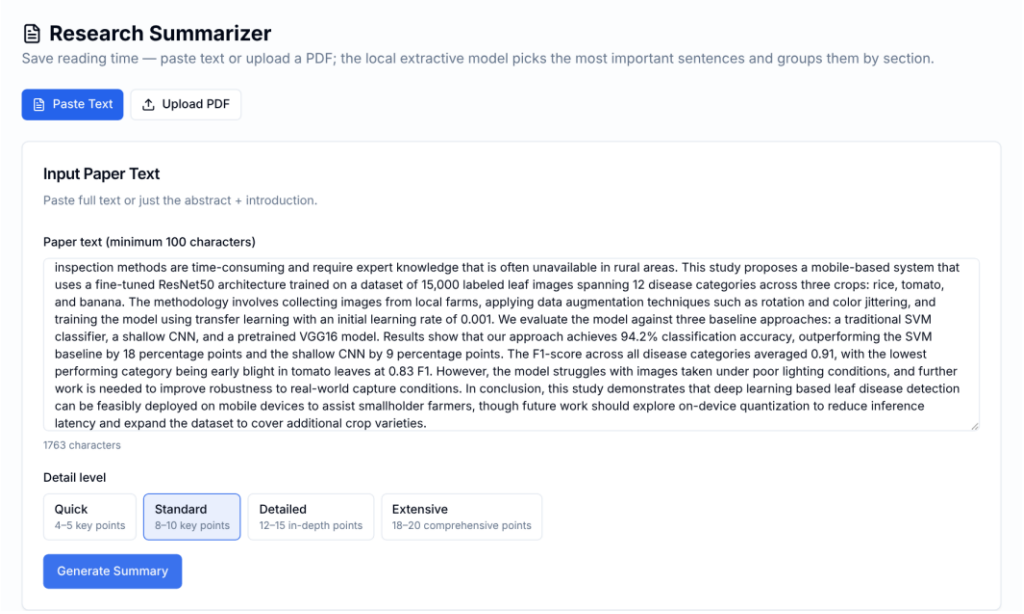


Figure C.1: Summariser input with detail-level presets

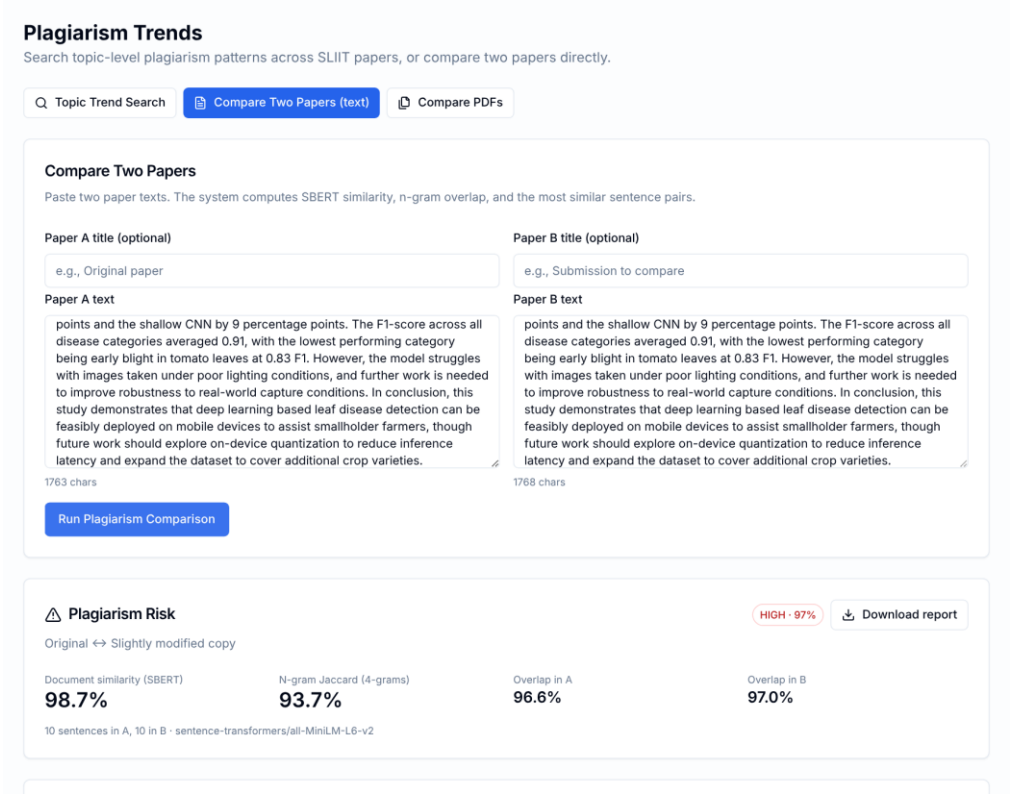


Figure C.2: Compare Two Papers input view

Appendix D: User Acceptance Testing Protocol

Participants complete the following tasks unaided; completion, time and errors are recorded and the System Usability Scale [43] is administered afterwards.

Table D.1: UAT tasks and recording sheet for Module 3

Task	Expected outcome	Completed (Y/N)	Time (s)	Errors / comments
T1 Summarise a paper PDF at the quick preset and download the report	Report saved			
T2 Categorise your abstract and open one related paper	Categories + RDA link			
T3 Compare your draft with a related paper and explain the risk level	Correct interpretation			
T4 Search plagiarism trends for your topic	Bucket and pairs shown			